제 3 장 반파 정류기

Half-wave Rectifier
AC→ DC

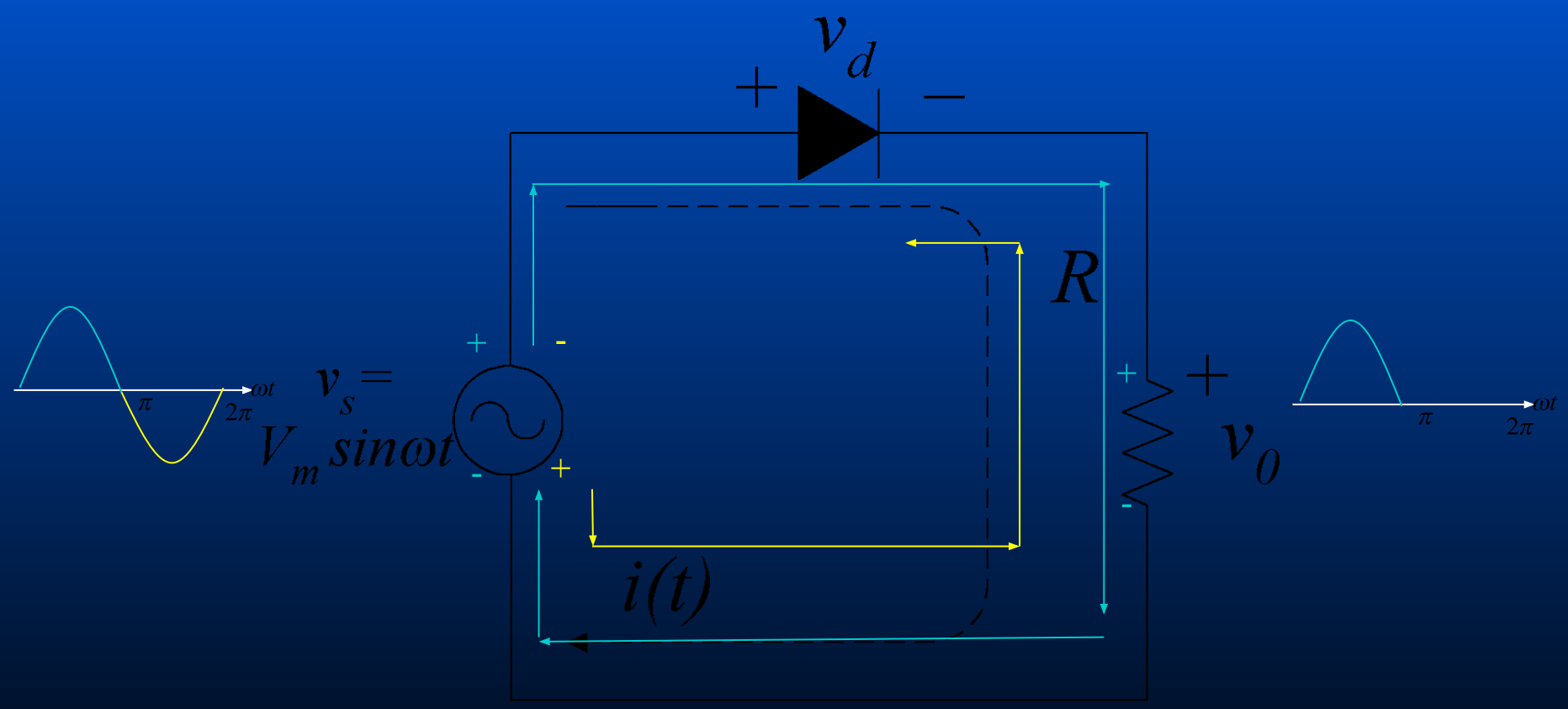
3장 반파정류회로의 구성

3.1 서론
3.2 저항부하(Resistive Load)
3.3 R-L 부하
3.4 OrCAD 시뮬레이션
3.5 R-L 기전력부하
3.6 인덕터 기전력부하
3.7 환류 다이오드(Freewheeling Diode)
3.8 R-C 반파정류회로
3.9 반파제어 정류기
3.10 OrCAD 시뮬레이션

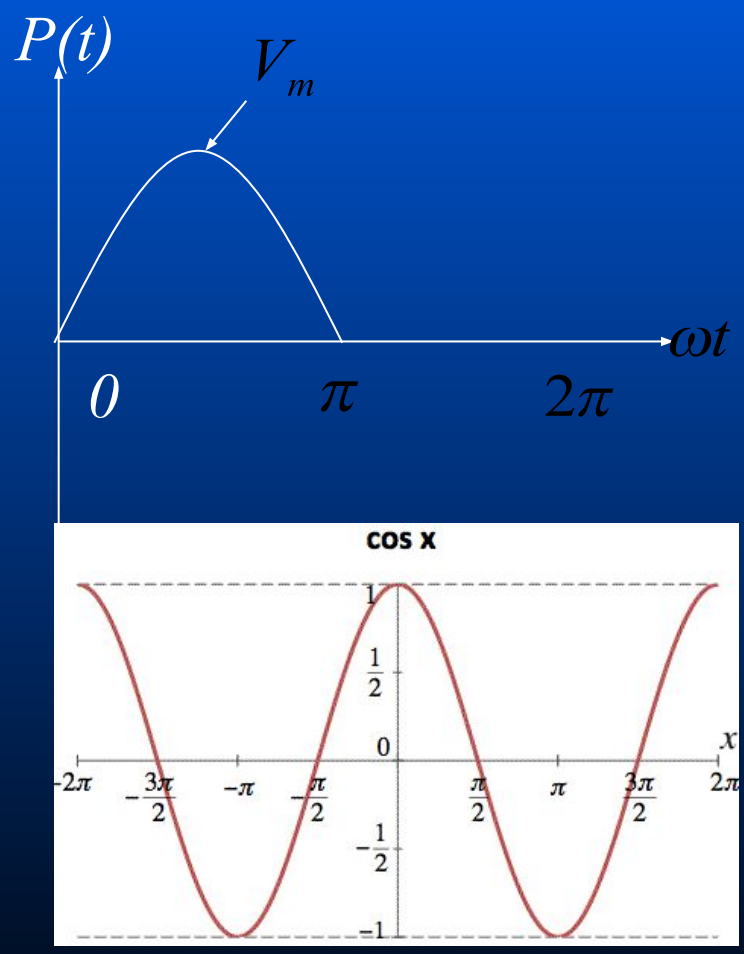
3.1 서론

- 정류(Rectification)
  AC→DC로 변환
- 반파 정류
  - 공급되는 전류 평균: 0이 아님
  - 변압기 성능에 문제점 유발 가능
  - 실제로는 전파 전류가 주로 이용
- OrCAD를 이용하여 계산: 가장 간편함

3.2 저항부하(Resistive Load)



3.2 저항부하: 반파의 평균값



- 전압의 평균값

$$V_0 = \frac{1}{2\pi}\int_0^{\pi} V_m \sin\omega t \, d(\omega t)$$

$$= \frac{V_m}{2\pi}\left[-\cos\omega t\right]_0^{\pi}$$

$$\int \sin dt = -\cos t$$

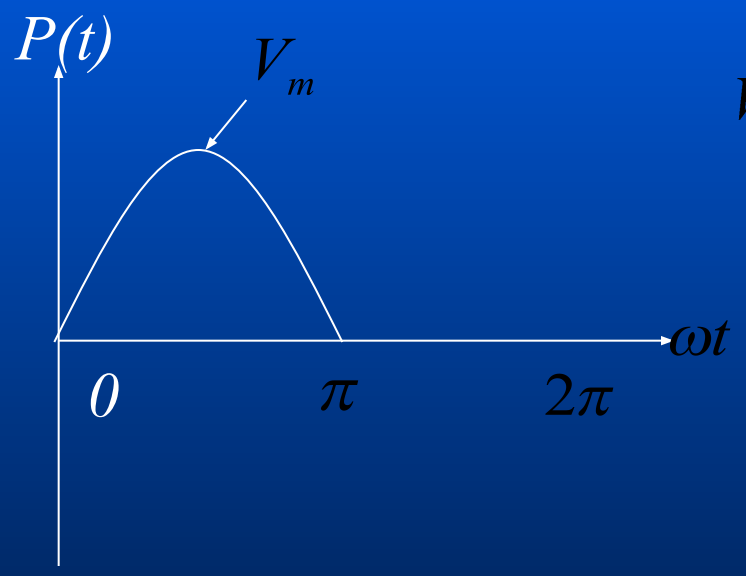
$$= \frac{V_m}{2\pi}\left(-\cos\pi - (-\cos 0)\right)$$

$$= \frac{V_m}{2\pi}\left(-(-1)+1\right) = \frac{V_m}{\pi}$$

- 전류의 직류 성분

$$I = \frac{V_0}{R} = \frac{V_m}{\pi R}$$

3.2 저항부하: 반파의 실효값



$$V_{rms} = \sqrt{\frac{1}{2\pi}\int_0^{\pi} (V_m \sin\omega t)^2 \, d(\omega t)}$$

$$\left(\sin^2 \omega t = \frac{1-\cos 2\omega t}{2} \right)$$

$$= \sqrt{\frac{V_m^{\,2}}{2\pi}\left[\frac{\omega t}{2} - \frac{\sin 2\omega t}{4}\right]_0^{\pi}}$$

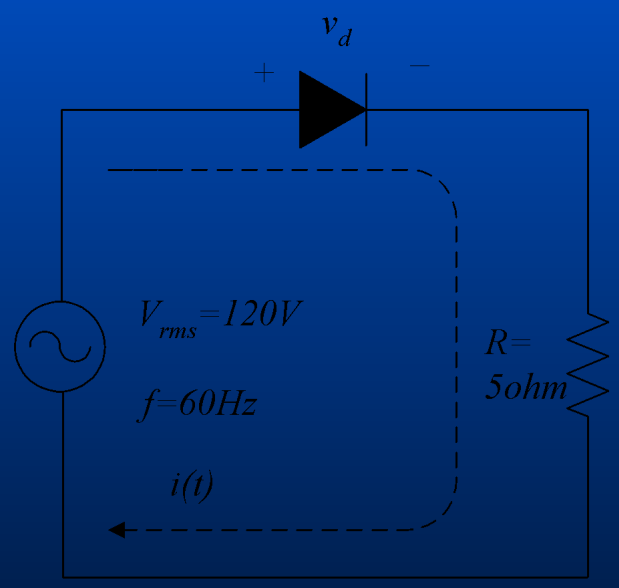
$$= \sqrt{\frac{V_m^{\,2}}{2\pi}\left(\frac{\pi}{2} - \cancel{\frac{\sin 2\pi}{4}} - \cancel{\frac{0}{2}} + \cancel{\frac{\sin 0}{4}}\right)} = \frac{V_m}{\sqrt{2}}$$

- 전류의 실효값

$$I_R = \frac{V_{rms}}{R} = \frac{V_m}{2R}$$

예제 3-1 저항부하를 가진 반파정류기(1)

(a) 평균 전류



$$V_0 = \frac{1}{2\pi}\int_0^{\pi} V_m \sin\omega t \, d(\omega t)$$

$$= \frac{1}{2\pi}\int_0^{\pi} \sqrt{2}(120)\sin\omega t \, d(\omega t)$$

$$= \frac{\sqrt{2}(120)}{2\pi}\left[-\cos\omega t\right]_0^{\pi}$$

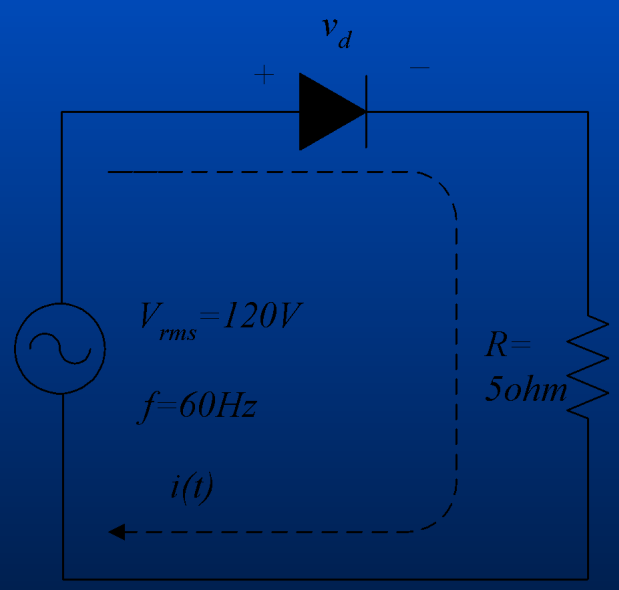
$$= \frac{\sqrt{2}(120)}{2\pi}(-\cos\pi + \cos 0)$$

$$= \frac{\sqrt{2}(120)}{\pi} = 54.02[V]$$

$$\therefore I = \frac{V_0}{R} = \frac{54.02}{5} = 10.8[A]$$

예제 3-1 저항부하를 가진 반파정류기(2)

(b) 실효 전압



$$V_{rms} = \sqrt{\frac{1}{2\pi}\int_0^{\pi} (V_m \sin\omega t)^2 \, d(\omega t)}$$

$$\left(\sin^2 \omega t = \frac{1-\cos 2\omega t}{2} \right)$$

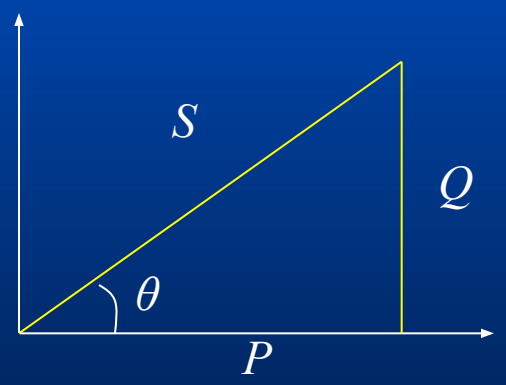
$$= \sqrt{\frac{V_m^{\,2}}{2\pi}\left[\frac{\omega t}{2} - \frac{\sin 2\omega t}{4}\right]_0^{\pi}}$$

$$= \sqrt{\frac{V_m^{\,2}}{2\pi}\left(\frac{\pi}{2} - \frac{\sin 2\pi}{4} - \frac{0}{2} + \frac{\sin 0}{4}\right)}$$

$$= \frac{\sqrt{2}(120)}{2} = 84.9V$$

예제 3-1 저항부하를 가진 반파정류기(3)

(c) 역률

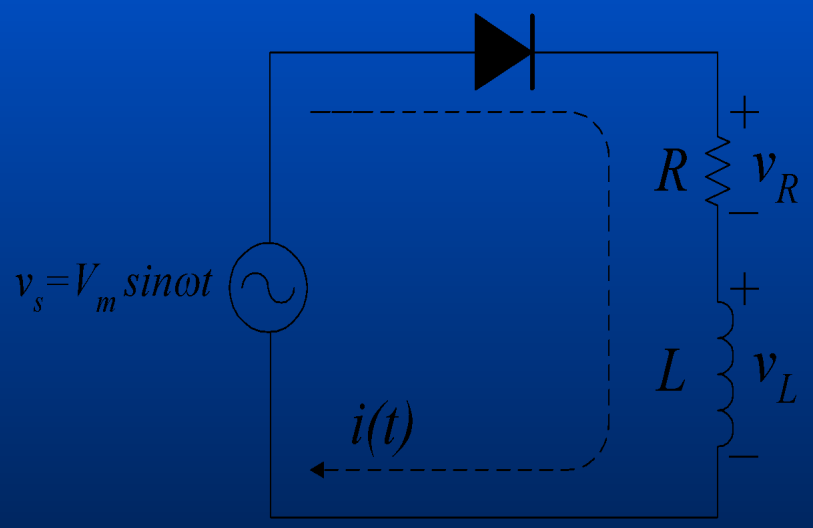


$$P = \frac{V_{rms}{}^2}{R} = \frac{84.9^2}{5} = 1440W$$

$$I_{rms} = \frac{V_m}{2R} = \frac{\sqrt{2}(120)}{2(5)} = 17\,A$$

$$pf = \frac{p}{V_{rms} I_{rms}} = \frac{1440}{(120)(17)} = 0.707$$

3.3 R-L 부하



- By KVL

$$L\frac{di}{dt} + Ri = V_m \sin\omega t$$

$$\omega L\frac{di}{d\omega t} + Ri = V_m \sin\omega t$$

3.3 R-L 부하: 전류계산(1)

$$\omega L \frac{di}{d\omega t} + Ri = V_m \sin \omega t \qquad \boxed{\times \frac{1}{\omega L}}$$

$$\frac{di}{dt} + \frac{R}{\omega L} i = \frac{V_m}{\omega L} sin\, \omega t \qquad \boxed{\times e^{\frac{R}{\omega L}\omega t}}$$

$$\frac{di}{d\omega t} e^{\frac{R}{\omega L}\omega t} + \frac{R}{\omega L} e^{\frac{R}{\omega L}\omega t} i = \frac{V_m}{\omega L} e^{\frac{R}{\omega L}\omega t} sin\, \omega t$$

$$\frac{d}{d\omega t}\left(ie^{\frac{R}{\omega L}\omega t} \right) = \frac{V_m}{\omega L} e^{\frac{R}{\omega L}\omega t} sin\, \omega t d\omega t \qquad \boxed{d(A \cdot B) = dA \cdot B + A \cdot dB}$$

$$ie^{\frac{R}{\omega L}\omega t} = \int \frac{V_m}{\omega L} e^{\frac{R}{\omega L}\omega t} \sin \omega t d\omega t + A$$

3.3 R-L 부하: 전류계산(2)

$$ie^{\frac{R}{\omega L}\omega t} = \int \frac{V_m}{\omega L} e^{\frac{R}{\omega L}\omega t} \sin\omega t d\omega t + A$$

$$\times e^{-\frac{R}{\omega L}\omega t}$$

$$i = e^{-\frac{R}{\omega L}\omega t} \frac{V_m}{\omega L} \int e^{\frac{R}{\omega L}\omega t} \sin\omega t d\omega t + Ae^{-\frac{R}{\omega L}\omega t}$$

$$\int e^{au} \sin bu du = \frac{e^{au}(a\sin bu - b\cos bu)}{a^2 + b^2}$$

$$i = e^{-\frac{R}{\omega L}\omega t} \frac{V_m}{\omega L} \frac{e^{\left(\frac{R}{\omega L}\right)\omega t}\left(\left(\frac{R}{\omega L}\right) sin\omega t - cos\omega t\right)}{\left(\frac{R}{\omega L}\right)^2 + 1^2} + Ae^{-\frac{R}{\omega L}\omega t}$$

3.3 R-L 부하: 전류계산(3)

$$i = e^{-\frac{R}{\omega L}\omega t}\frac{V_m}{\omega L}\frac{e^{\left(\frac{R}{\omega L}\right)\omega t}\left(\left(\frac{R}{\omega L}\right)\sin\omega t - \cos\omega t\right)}{\left(\frac{R}{\omega L}\right)^2 + 1^2} + Ae^{-\frac{R}{\omega L}\omega t}$$

$$= \frac{V_m}{\omega L}\frac{\left(\left(\frac{R}{\omega L}\right)\sin\omega t - \cos\omega t\right)}{\left(\frac{R^2 + (\omega L)^2}{(\omega L)^2}\right)} + Ae^{-\frac{R}{\omega L}\omega t}$$

$$= \frac{V_m}{\left(R^2 + (\omega L)^2\right)}\left(R\sin\omega t - \omega L\cos\omega t\right) + Ae^{-\frac{R}{\omega L}\omega t}$$

3.3 R-L 부하: 전류계산(4)

$$i = \frac{V_m}{\left(R^2 + (\omega L)^2\right)}\left(R\sin\omega t - \omega L\cos\omega t\right) + Ae^{-\frac{R}{\omega L}\omega t}$$

$$\boxed{a\cos x + b\sin x = \sqrt{a^2 + b^2}\,\sin(x + \delta),\ \delta = \tan^{-1}\frac{a}{b}}$$

$$i = \frac{V_m\sqrt{\left(R^2 + (\omega L)^2\right)}}{\left(R^2 + (\omega L)^2\right)}\sin(\omega t - \theta) + Ae^{-\frac{R}{\omega L}\omega t}$$

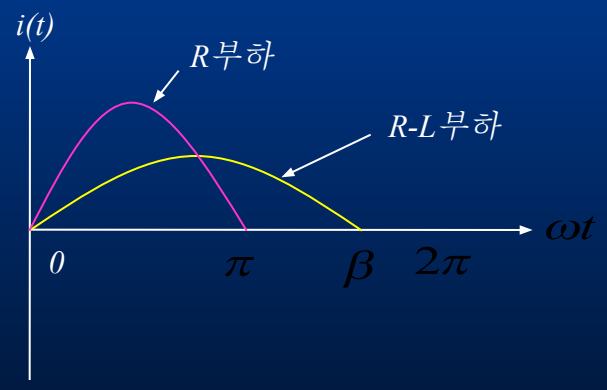
$$= \frac{V_m}{\sqrt{\left(R^2 + (\omega L)^2\right)}}\sin(\omega t - \theta) + Ae^{-\frac{R}{\omega L}\omega t} = \frac{V_m}{Z}\sin(\omega t - \theta) + Ae^{-\frac{R}{\omega L}\omega t}$$

$$where\ \ Z = \sqrt{\left(R^2 + (\omega L)^2\right)},\theta = \tan^{-1}\frac{\omega L}{R}$$

3.3 R-L 부하: 전류계산(5)

$$i = \frac{V_m}{Z}\sin(\omega t - \theta) + Ae^{-\frac{R}{\omega L}\omega t}$$

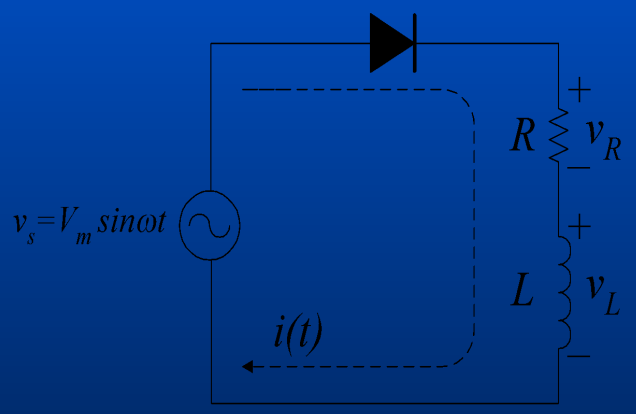
- A를 구하기 위한 초기조건 $\omega t=0$ 일 때 $i=0$



$$Ae^{-0} = 0 - \frac{V_m}{Z}\sin(0-\theta) = \frac{V_m}{Z}\sin(\theta)$$

$$\therefore i = \frac{V_m}{Z}\left[\sin(\omega t - \theta) + \sin\theta\, e^{-\frac{R}{L}t}\right]$$

3.3 R-L 부하: 정상과 과도



- By KVL

$$L\frac{di}{dt} + Ri = V_m \sin\omega t$$

- 전류값

$$i(t) = \frac{V_m}{Z}\left[\sin(\omega t - \theta) + \sin\theta \cdot e^{-\frac{R}{L}t}\right]$$

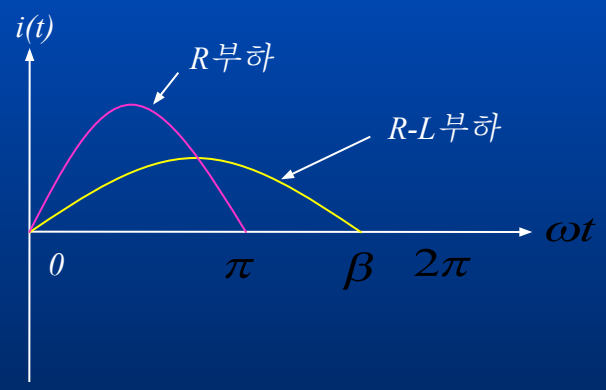
- 전류=정상치+과도치

$$i(t) = i_f(t) + i_n(t)$$

*여기서 $i_f = \frac{V_m}{Z}\sin(\omega t - \theta),\ i_n = \frac{V_m}{Z}\sin\theta e^{-\frac{R}{L}t}$

3.3 R-L 부하: 소호각 계산

- β: 소호각(Extinction Angle)



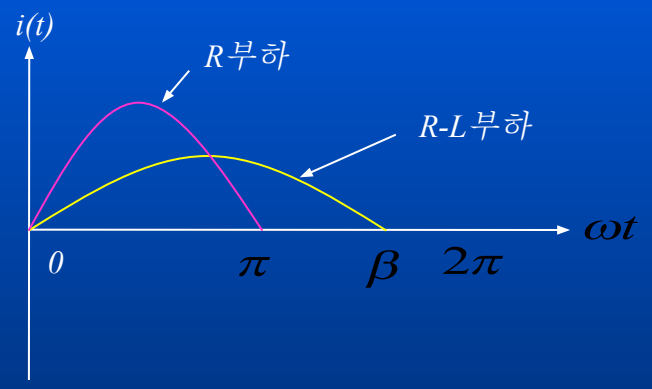
$$i(t) = \frac{V_m}{Z}\left[\sin(\omega t - \theta) + \sin\theta \cdot e^{-\frac{R}{L}t}\right]$$

$$i(\beta) = 0 = \frac{V_m}{Z}\left[\sin(\beta t - \theta) + \sin\theta \cdot e^{-\frac{R\beta}{L\omega}t}\right]$$

$$\therefore \sin(\beta t - \theta) + \sin\theta \cdot e^{-\frac{R\beta}{L\omega}t} = 0$$

- β의 계산
  - 수치기법(Numerical Analysis) 이용계산
  - OrCAD를 이용하여 계산: 가장 간편함

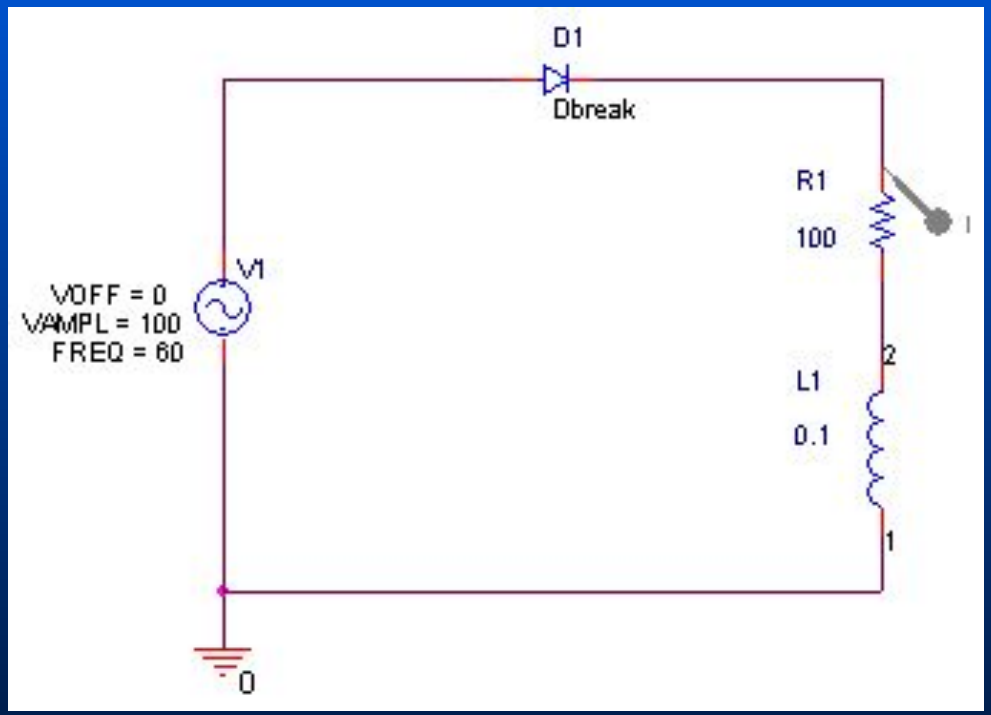
3.3 R-L 부하: 종합



$$i(\omega t) = \begin{cases} \dfrac{V_m}{Z}\left[\sin(\omega t - \theta) + \sin\theta \cdot e^{-\frac{R}{L}t}\right] & at\ 0 \le \omega t \le \beta \\ 0 & at\ \beta \le \omega t \le 2\pi \end{cases}$$

*여기서 $Z = \sqrt{R^2 + (\omega L)^2}$, $\theta = \tan^{-1}\left(\dfrac{L}{R}\right)$

예제 3-3 R-L 부하를 가진 반파정류기



- *Parameter*
  - ✔ $R=100\ \Omega$
  - ✔ $L=0.1\ H$
  - ✔ $\omega=377\ rad/s$, $f=\omega/2\pi=60\ Hz$
  - ✔ $V_m=100\ V$
  - ✔ *Transient:*

    0 0.01m 50ms